

# AskIt Research Lab — Home Assignment Submission

Repository (code, data, full logs, commit history): <https://github.com/DeanData/askit-persona-function>

Metric, committed before any text was generated:

<https://github.com/DeanData/askit-persona-function/blob/main/METRIC.md>

**Part 3 (slow research) status:** not yet completed at the time of this submission. See REPORT.md's Part 3 section below for the current status; if completed later in the 5-hour budget, the repository's REPORT.md will reflect the finished version even if this PDF was generated earlier.

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*Full reasoning, dead ends, and every decision's rationale live in WORKLOG.md (chronological) and METRIC.md (the metric, committed before any generation). This report summarizes; those are the record.*

## The function

`f(audience description, persona description, context) -> text`. Audiences: 5 finance- forum populations (`r/personalfinance`, `r/wallstreetbets`, `r/fatFIRE`, `r/thetagang`, `r/povertyfinance`), chosen over demographic-bin alternatives because they give a genuine two-attribute intersection (risk-orientation × sophistication) without inventing one — `fatFIRE` is the dev intersectional audience, `thetagang` (held out) mirrors its compound pattern but swaps the risk attribute. `povertyfinance` is held out as a single-attribute, demographically distinct population. Personas: a structured schema (age, age-conditioned occupation, region, family status, one open-ended detail), rendered through one fixed template, 40 per audience, fixed seed, shared roster across every condition (paired comparison). No persona field restates the audience's defining trait. Contexts: one structural ("\$10k advice-giving," forces justification/argument) and one surface ("weekly money check-in," low-stakes, lets register surface unforced) — deliberately chosen to pull on the function differently, both finance-anchored (checked against the generality requirement; audiences, not topics, are what must generalize).

## Metric (committed in METRIC.md before any text was generated)

Ten cheap stylometric features (6 structural: sentence length, lexical diversity, word length, first-person rate, hedge rate, punctuation density; 4 surface: emoji, exclaim, all-caps, meme-token rate) — validated empirically in a Phase 1 pre-check on real data (5-class nearest-centroid accuracy 42.0% full vs. 41.6% structural-only vs. 20% chance: separation is structural, not slang-driven). Ground truth is real Reddit text, split into a disjoint eval pool (242/audience, used only for reference statistics) and a few-shot pool (8/audience, used only in prompts) — a contamination risk caught before it was built, not after. Success is defined as **matching real audiences, not maximizing separation**: an earlier draft treated a bigger between/within ratio as better, which directly rewards within-audience collapse; both fidelity measures are now signed relative error against real references (0 = perfect match). Self-labeled texts (literal audience name-dropping) are excluded from fidelity, counted separately. Bootstrap 95% CIs (n=1000) on every number. **What a perfect score would still miss**: style, not content; `personalfinance`/`povertyfinance` are known-hard to separate on this feature set; the meme-token wordlist is crude; personas are demographically uniform by design, so some fidelity gap may reflect demographic mismatch rather than voice failure.

## Numbers (see [artifacts/baseline\\_vs\\_fewshot.png](#))

**Baseline** (naive prompt: audience + persona + context, nothing else) shows a consistent homogenization pattern on dev: within-variation fidelity negative for all 3 dev audiences (`personalfinance` −0.17, `wallstreetbets` −0.43, `fatFIRE` −0.12; all CIs exclude zero), centroid drift substantial (0.37–0.52), between-separation also negative (−0.20). The baseline is more homogeneous than real text on every axis at once.

**Fewshot** (8 real per-audience examples added as a style primer) was evaluated against this on the same paired roster. On dev, the result is **mixed-to-negative**: within- variation flat in aggregate (MARE 0.242→0.243) and worse for fatFIRE specifically (−0.12→−0.15); between-separation moved away from zero (−0.20→−0.25, worse); centroid drift improved only marginally, driven almost entirely by fatFIRE. On held-out (thetagang, povertyfinance — run exactly once, after dev was finalized), the picture flips: every measure moves in the favorable direction (within-variation, centroid drift, and between-separation all improve). But baseline/fewshot bootstrap CIs overlap for most held-out measures — a consistent directional signal, not a statistically decisive win. **Dev and held-out disagreed about whether the improvement helped** — exactly the disagreement the held-out mechanism exists to surface.

## What didn't work

Few-shot's dev self-labeling rate more than doubled (6/240→13/240 texts), concentrated almost entirely in fatFIRE (3→12 of 80). Inspected the fatFIRE few-shot pool directly: 2 of its 8 real examples literally self-label ("Fatfire in unconventional cities," "... achieve fatFIRE?"). Showing the model real text that names its own community taught it to imitate that. This is a real, well-evidenced negative finding, not just a weak result. **Not fixed now**: filtering self-labeled examples from the few-shot pool and rerunning is exactly the "further tuning past the first defensible improvement" we agreed to cut if time was short — named here as a next step, not executed.

## Part 3: Slow research

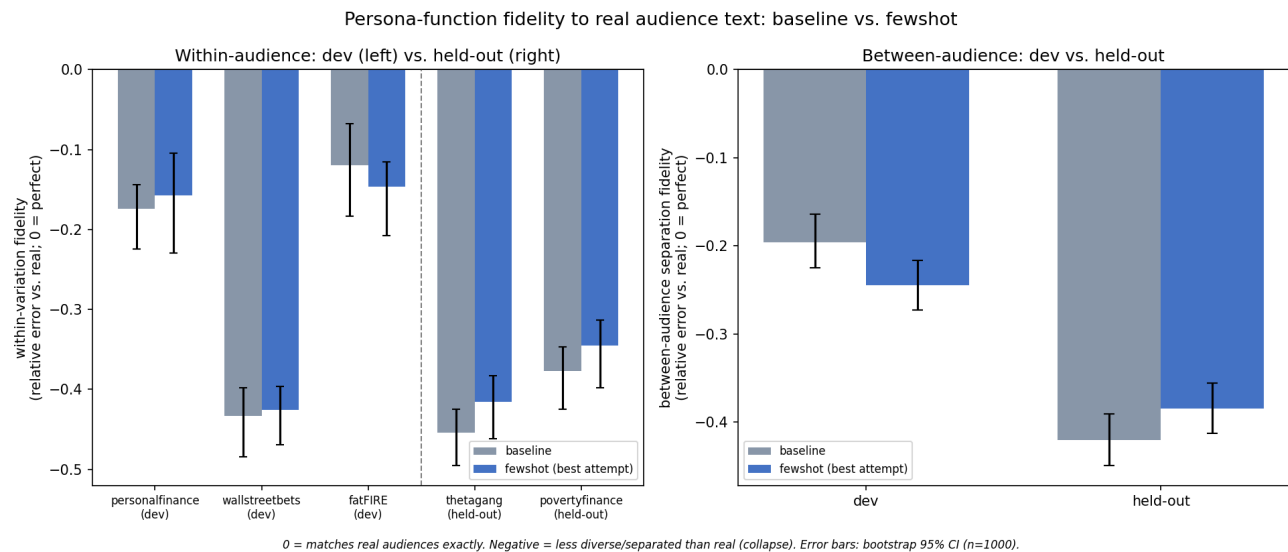
*Placeholder — pending remaining time in the 5-hour budget. Working through the mechanism, cost, falsification test, and kill criterion collaboratively rather than proposing it cold; to be filled in if time allows, reported honestly as not reached otherwise.*

## What we'd do with another week

Filter self-labeled examples from the few-shot pool and rerun, to isolate whether the between-separation regression was a few-shot-mechanism problem or specifically the self-labeling leakage. Add a third, topic-neutral elicitation context to test whether audience voice survives outside finance-adjacent situations — the current two contexts can't distinguish that. Relax the demographically-uniform persona design (named as a blind spot in METRIC.md) to test whether real audiences' demographic differences, not just voice, are driving some of the fidelity gap. Re-run the classifier-style separation check from Phase 1 directly on generated text (not just real text) with cross-validation instead of a single split, now that we have a reason to trust it at this sample size. Widen the audience set beyond one dataset (investing forums) to check whether the metric and findings generalize to a genuinely different domain.

*The required paragraph on where Claude Code was plausible but wrong now lives in its own file, CLAUDE\_CODE\_WAS\_WRONG.md, as a distinct deliverable.*

## Required visual



## Where Claude Code was plausible but wrong

Early in Part 1, I proposed a 2x2 design (risk-orientation × sophistication) for the five audiences and labeled fatFIRE as "the intersectional audience." This sounded coherent — fatFIRE genuinely is defined by two attributes — but was actually inconsistent: in a full 2x2, every audience (personalfinance, wallstreetbets, fatFIRE, thetagang) is equally defined by both attributes, so none is uniquely "the" compound one. The design silently erased the single-vs-compound contrast the requirement ("at least one audience must be an intersection... rather than a single attribute") actually depends on — "intersectional" would have been true by construction, not by design. The candidate caught it by asking directly: "isn't by your logic the others also an intersection in the same way?" The fix was to make risk-orientation the sole defining attribute for personalfinance/ wallstreetbets/povertyfinance (sophistication left unspecified, matching the guidelines' own worked example), reserving the deliberate two-attribute combination for fatFIRE (dev) and thetagang (held out, testing whether the compound pattern generalizes rather than being memorized from fatFIRE alone). Logged in `WORKLOG.md` the moment it was caught.

## Audience descriptions

Population-level descriptions fed to `f()` as one of its two inputs. Full file: [data/audiences.json](#).

### personalfinance (dev)

Members of the `r/personalfinance` community: people seeking or giving advice on everyday financial situations, including budgeting, debt, saving, home-buying, and insurance. Conservative risk orientation — favors safety, index funds, emergency funds, and caution over speculation. Financial sophistication is mixed — includes both first-time budgeters and experienced planners; not a defining trait of this audience.

### wallstreetbets (dev)

Members of the `r/wallstreetbets` community: people who trade or discuss trading as high-risk, high-reward speculation, including options, meme stocks, and leveraged bets. Risk-tolerant orientation — comfortable with, and often seeking out, high volatility and large swings. Financial sophistication is mixed — includes true novices alongside experienced traders; not a defining trait of this audience.

## fatFIRE (dev\_intersectional)

Members of the r/fatFIRE community: high-income earners pursuing early retirement at a substantially above-average spending level, through saving, diversified investing, and tax optimization. Conservative risk orientation — prioritizes capital preservation and diversification over speculation, despite high income. Financial sophistication is advanced — assumes familiarity with tax-advantaged accounts, asset allocation, and long-horizon planning; this conservative-and-advanced combination is what distinguishes this audience from personalfinance.

## thetagang (heldout\_intersectional)

Members of the r/thetagang community: active options traders focused on selling options premium — covered calls, cash-secured puts, credit spreads — as a recurring income strategy. Risk-tolerant orientation — comfortable holding directional and assignment risk for premium income. Financial sophistication is advanced — assumes working knowledge of options mechanics and position management; this risk-tolerant-and-advanced combination mirrors fatFIRE's compound pattern but swaps the risk attribute.

## povertyfinance (heldout)

Members of the r/povertyfinance community: people managing financial precarity, including living paycheck to paycheck and navigating debt, benefits, or unstable income. Conservative risk orientation, driven by constraint and necessity rather than choice; caution here reflects survival margin, not risk preference in the abstract. Financial sophistication is mixed and not a defining trait of this audience.

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## Persona descriptions (sample)

One individual drawn from an audience, the other of f()'s two inputs. Structured schema (age, age-conditioned occupation, region, family status, one open-ended detail) rendered through a fixed template — full design rationale in WORKLOG.md. 40 personas per audience, 200 total, fixed seed, shared roster across every condition. Full roster: [data/personas/](#). Two samples below, one per risk-orientation:

- **personalfinance:** 57, engaged, lives in Seattle, Washington, works as a nurse practitioner. Recently adopted a rescue dog.
- **wallstreetbets:** 31, divorced, shares custody of one child, lives in a small town in rural Minnesota, works as a physical therapist. Started taking pottery classes recently.